Exercise 2

Goal: implement k-means clustering using the classic Lloyd iterative refinement scheme.

- *M* points in 2D space, distributed across *N* chares in a **chare array**
- each Lloyd iteration is one **broadcast** + two **reductions** over a chare array.
- Charm4Py **futures** are used as reduction targets

Algorithmic details

1. main() generates K random points in [0, 1)^2 as initial centroid guesses.
2. **TODO: main() broadcasts the K centroids to every Points chare via the assign() entry method.**
3. Each chare assigns each of its local points to the nearest centroid (by squared Euclidean distance)
4. **TODO: each chare then contributes to two sum reductions:**
   - counts -- length-K integer array, number of local points assigned to each cluster,
   - coords -- length-2K double array, sum of x and y coordinates of local points assigned to each cluster (interleaved: x_0, y_0, x_1, y_1, ...).
5. **TODO: main() receives both reduced arrays via futures.**
6. Centroids are updated.
7. Convergence: when no centroid coordinate moves by more than THRESHOLD (0.001) between successive iterations, stop. Otherwise, repeat.